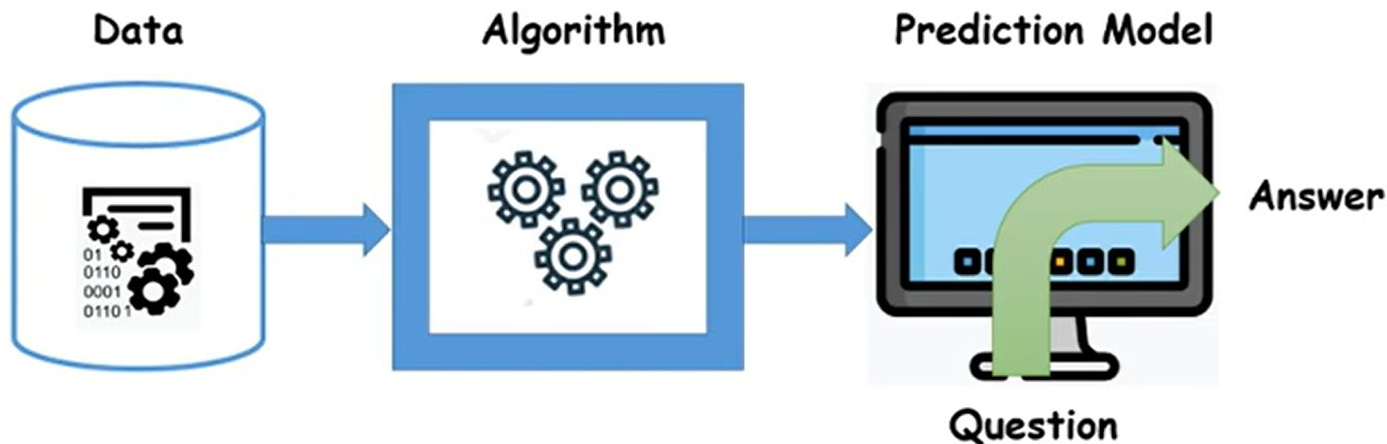


Neuro Fuzzy System

CSE 4237
Soft Computing

Modeling of Systems

What is machine learning?



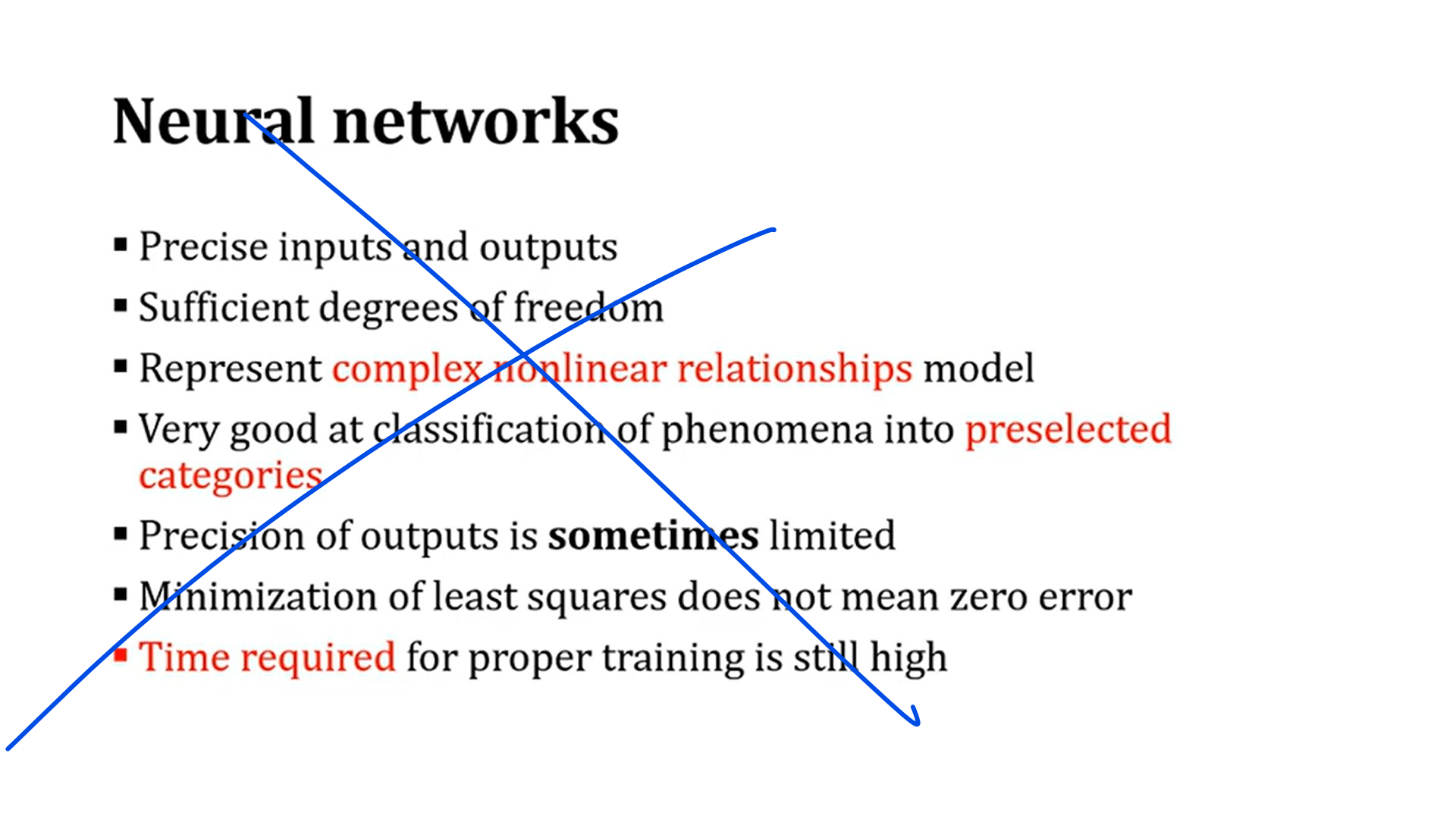
Why we need neural networks and fuzzy logic systems?

- In data and/ or information processing, the objective is generally to **gain an understanding** of the phenomena involved and to **evaluate relevant parameters quantitatively**.
- This is usually accomplished through “**modeling**” of the systems.
 - Either **experimentally or analytically** (using **mathematics** and **physical principles**)
- Once we have a model of system, we can carry out various procedures (for example, **sensitivity analysis, statistical regression**, etc.).

Why we need neural networks and fuzzy logic systems?

- However, many situations are very complex and not well understood.
 - Physical measurements of the pertinent quantities are very difficult and expensive.
- These difficulties lead us to use of neural networks and fuzzy logic systems as a way of **obtaining models based on experimental measurements.**

Neural networks

- Precise inputs and outputs
 - Sufficient degrees of freedom
 - Represent **complex nonlinear relationships** model
 - Very good at classification of phenomena into **preselected categories**
 - Precision of outputs is **sometimes** limited
 - Minimization of least squares does not mean zero error
 - **Time required** for proper training is still high
- 

Fuzzy Systems

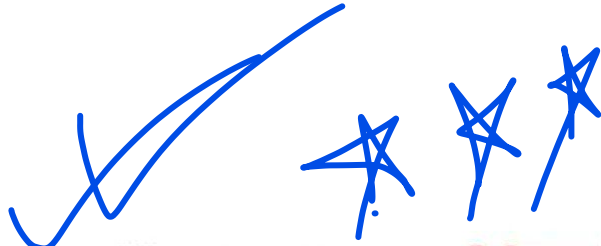
- * ✓ Reverse situations prevails compare to neural networks
- ✓ The **input and output variables** are encoded in **fuzzy representations**
- ✓ Interrelationships take the form of well-defined *if/then* rules

Fuzzy systems - Advantages

- Address the impression of the input and output variables
 - Use fuzzy numbers and fuzzy sets that can be expressed in **linguistic terms** (e.g., cold, warm, and hot)
- Greater flexibility in formulating system descriptions
 - **Complex process behavior can be described in general terms**
- Fuzzy description are more parsimonious, easy to formulate and modify

Neural networks and fuzzy logic systems

- ✓ Neural networks and fuzzy logic systems represent two **distinct** methodologies that deal with **uncertainty**.
 - These uncertainties usually arise from system complexity (complexity as a property of system description)
- ✓ Accomplish same results in different ways
 - Can speed the unraveling and specifying the mathematical models
 - Can be used to control nonlinear system
 - Perform **mapping with some degree of impression**



However, **combination of the two technologies** with the goal of **gaining the advantages of both** that is the focus of this soft computing course.

Neurofuzzy systems

- An integration of fuzzy logic and neural networks

□ Advantages of neurofuzzy systems

- Addressing the problems of large complex systems.
- Ability to perform **many-to-many** mappings
- Dealt with in a flexible, reliable, and near-optimal manner.

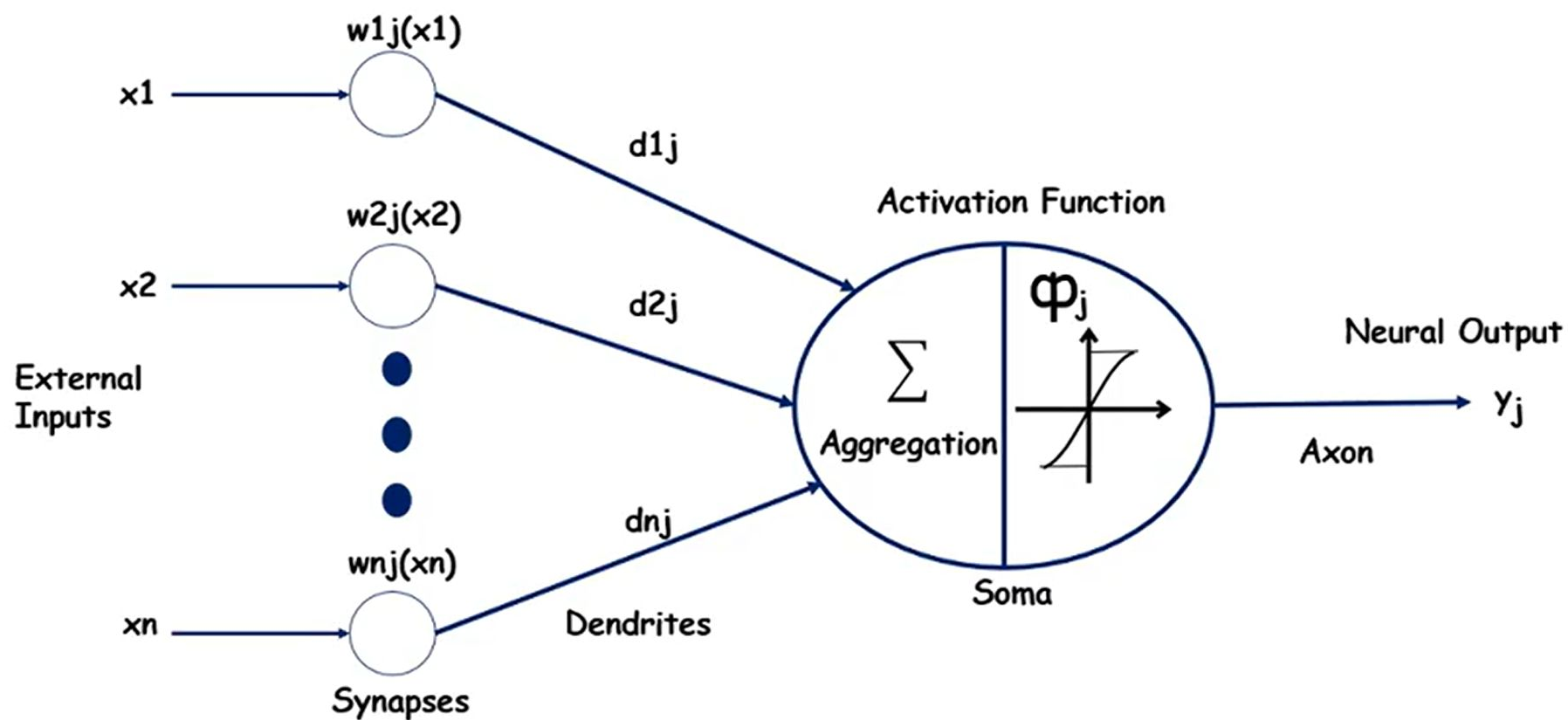
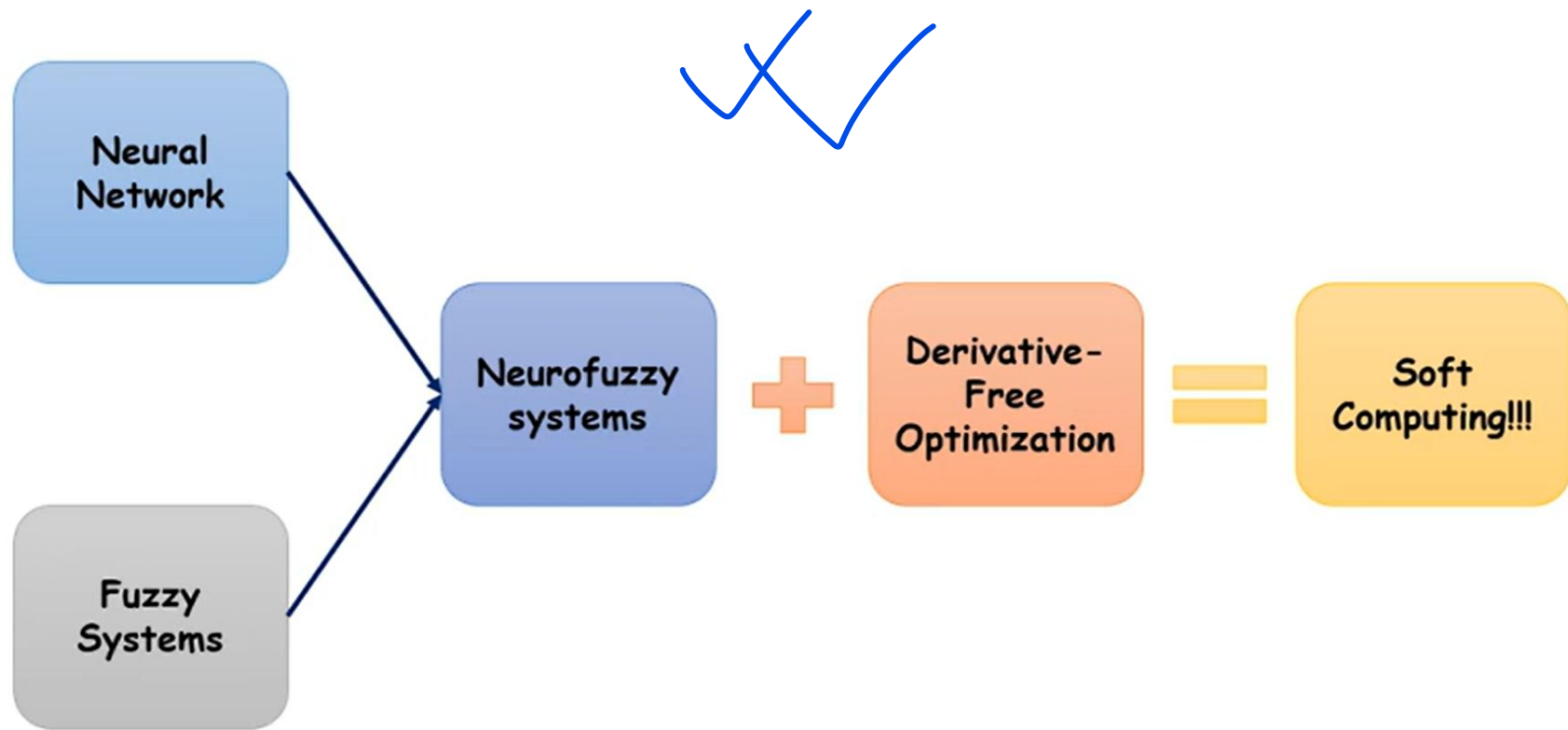
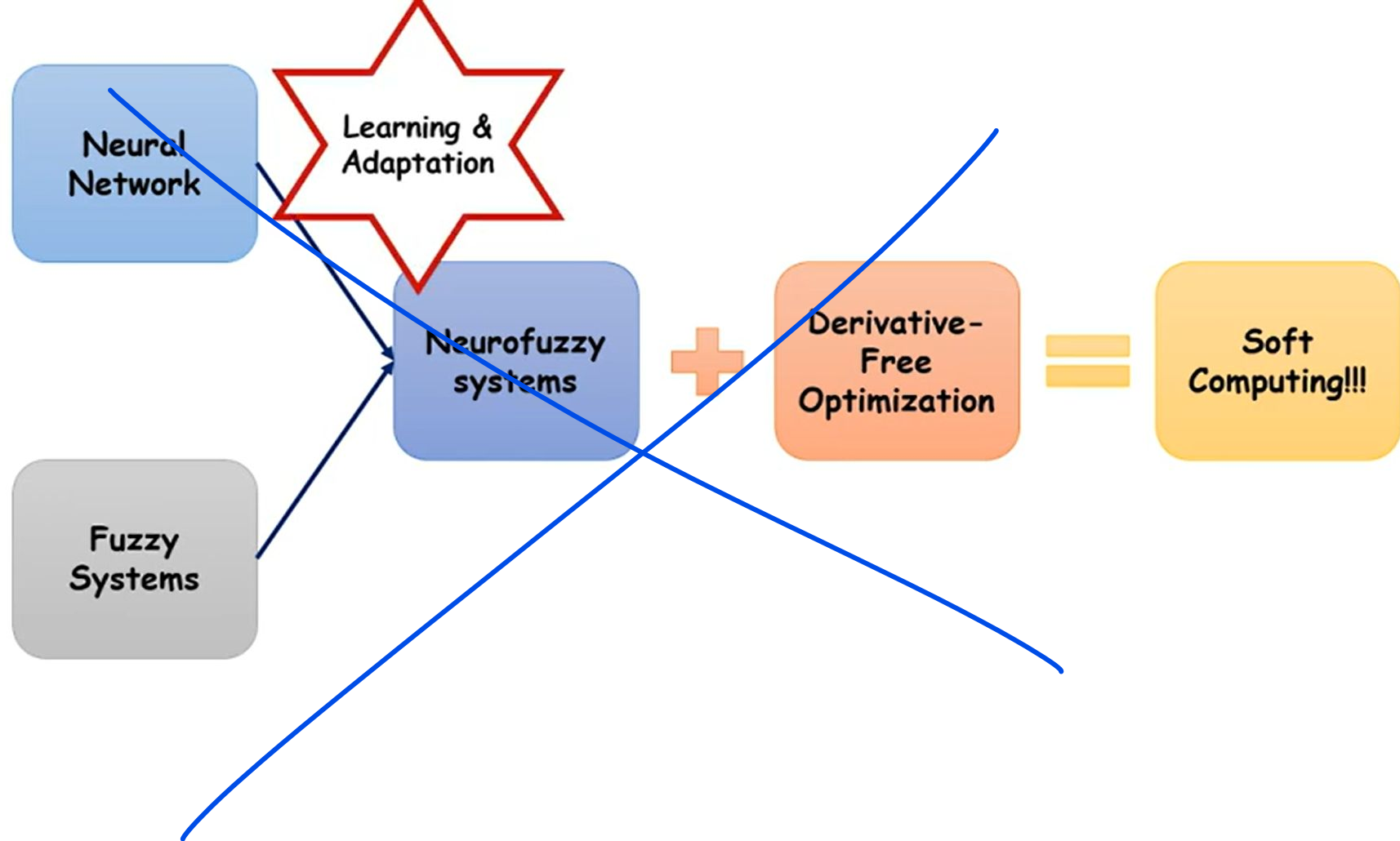
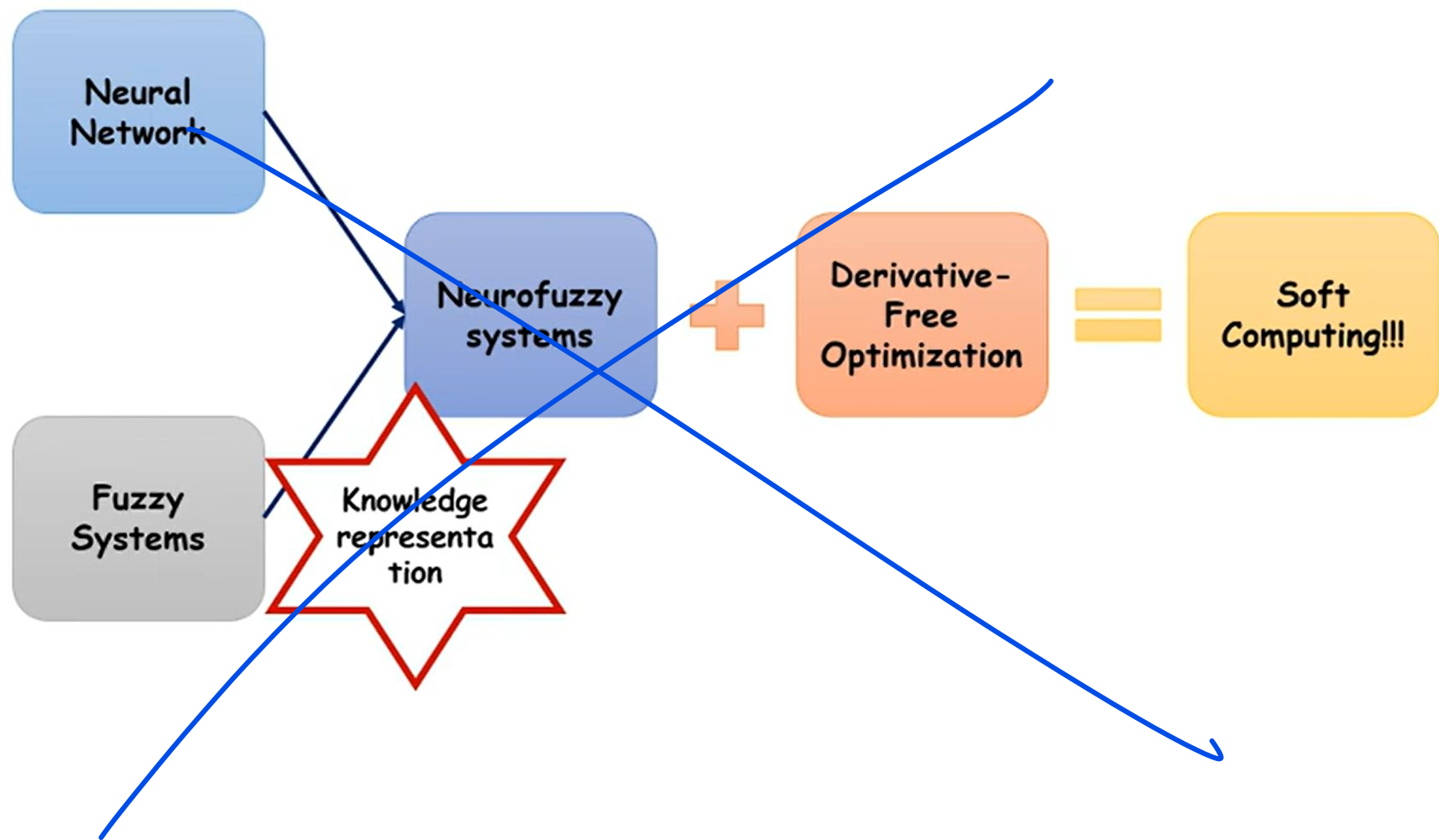


Figure: Simplified model of a neuron as an information processor.







Neural
Network

Neurofuzzy
systems



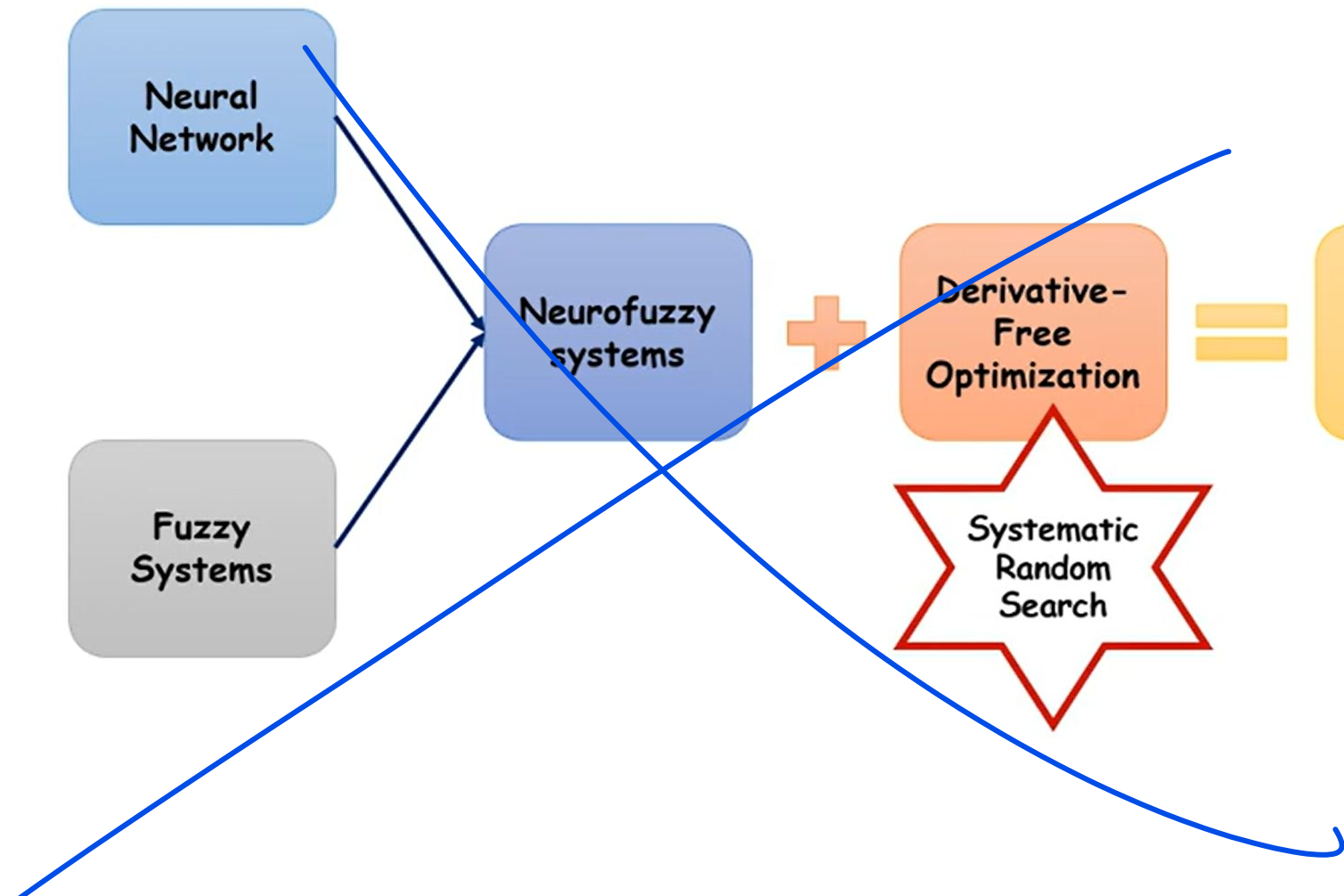
Derivative-
Free
Optimization



Soft
Computing!!!

Fuzzy
Systems

Systematic
Random
Search



Soft Computing

❑ Can be characterized as:

- ✓ Human expertise (fuzzy if/then rules)
- ✓ Biologically inspired computing models (NN)
- ✓ Optimization techniques (GA, SA, etc.)
- ✓ Numerical computation